

Probability before Kolmogorov

Section 5 · Live writing edition

Soma S Dhavala · IIT Jammu

2026-10-04

What does a probability mean?

In a hypothetical agent benchmark, success was recorded on 90 of 100 tasks. What does the statement “success probability is 0.9” assert, and what does it assume?

The calculus and its interpretation

Separate two questions: which probability calculations are valid, and what do the numbers represent?

A large rectangular area with a light gray dotted grid pattern, intended for taking notes or drawing.

Laplace's classical rule

A fair die has six equally likely faces. What probability does the classical rule assign to an even face?

A large grid of dots for writing the answer, consisting of 20 rows and 60 columns of small, light blue dots.

The assumption behind counting

What justifies assigning the same probability to every face?

A large rectangular area filled with a uniform grid of small, light gray dots, intended for taking notes or writing an answer.

A circular definition

Can “equally possible” define probability without already using a probability idea?

A large rectangular area filled with a light gray dot grid, intended for taking notes.

One experiment, two partitions

Toss a fair coin twice. What is the probability of at least one head? First list the ordered outcomes.

The same event with three cases

A three-case account lists: a head on the first toss; a tail then a head; two tails. Are these cases equally likely?

Partition dependence

What failed in the three-case calculation?

When the counting rule is valid

State a condition under which $P(A) = |A|/|\Omega|$ is justified for finite Ω .

A large grid of dots for writing the answer.

A measured fraction is not a probability

In n observed trials, the event A occurred $N_n(A)$ times. What does $N_n(A)/n$ describe?

The frequency interpretation

How does the frequency account define the probability of A ?

A large rectangular area filled with a grid of small, light blue dots, intended for taking notes or drawing diagrams.

What is a collective?

Why did von Mises impose a condition on selected subsequences?

A large rectangular area with a light gray dotted grid pattern, intended for handwritten notes.

Limits of the frequency account

Can a finite run establish that the limiting relative frequency exists? What about a one-time event?

A large rectangular area with a light gray dotted grid pattern, intended for taking notes or drawing.

A theorem or a definition?

Under the frequency interpretation, is convergence a theorem? Under the axioms, what is its status?

A large grid of dots for taking notes.

Probability as degree of belief

How might a probability describe uncertainty about an event that will not be repeated?

A large rectangular area with a light gray dotted grid pattern, intended for handwritten notes or a diagram.

Coherence constrains beliefs

What condition prevents a system of betting probabilities from guaranteeing a loss?

A large rectangular area with a light gray dotted grid pattern, intended for writing an answer.

Finite and countable additivity

Does coherence alone justify countable additivity?

A large rectangular area with a light gray dotted grid pattern, intended for writing or drawing.

Probability as a relation to evidence

Can probability be understood as a relation between evidence and a proposition, rather than a personal belief?

The principle of indifference returns

How does the logical account face the difficulty of assigning probabilities without enough evidence?

A large rectangular area filled with a grid of small, light blue dots, intended for taking notes or writing answers.

Fiducial probability

What was Fisher's fiducial argument intended to obtain?

A large rectangular area with a light gray dotted grid pattern, intended for a handwritten answer.

What fiducial probability is—and is not

Is fiducial probability an alternative interpretation of the probability calculus?

A large rectangular area filled with a grid of small, light blue dots, intended for taking notes or writing answers.

Bertrand: a question without a sampling rule

If someone says “choose a chord at random,” is the probability model specified?

A large grid of dots for taking notes.

What Bertrand does not explain

Does specifying a sampling rule provide a framework for proving results about infinite sequences or processes?

A large rectangular area filled with a grid of small, light blue dots, intended for taking notes or writing answers.

Measure theory preceded the axioms

Was measure theory invented to resolve Bertrand's paradox?

A large rectangular area with a light gray dotted grid pattern, intended for taking notes.

Infinite sequences need measurable events

To say that relative frequencies converge with probability one, what set must be an event?

A large rectangular area filled with a grid of small, light gray dots, intended for the user to write their answer to the question above.

Processes on function spaces

What kind of probability space can represent a random continuous path?

A large rectangular area filled with a grid of small, light gray dots, intended for taking notes or drawing.

Conditioning on a probability-zero event

Why does $P(A \mid B) = P(A \cap B)/P(B)$ not define conditioning when $P(B) = 0$?

A large rectangular area with a light gray dotted grid pattern, intended for a handwritten answer.

Axiomatization and mechanics

Was probability axiomatization also part of a broader mathematical programme?

A large grid of small dots, likely a placeholder for a diagram or a list of items.

What the axioms settle

Do the axioms determine which probability interpretation is correct?

A large rectangular area with a light gray dotted grid pattern, intended for taking notes.

Reading guide

Which arguments should be revisited in the main notes, and which belong to the Bertrand supplement?